

Preliminary report on the emotion classifier project

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Abstract

In this report we share our approach to the facial emotion recognition problem. The main focus of this paper is to give the reader a brief overview over the datasets, the data preprocessing and the model architecture used in our approach.

1. Introduction

First Section

2. Model Architectures

To address the facial emotion recognition task (FER), our group investigates multiple convolutional neural networks (CNNs) that have demonstrated strong performance in image classification and facial analysis tasks. CNNs build the base for most of our models. This decision is based on according literature emphasizing the clear performance advantage of CNNs compared to other approaches such as traditional machine learning models [1]. Each team member focuses on implementing and evaluating selected architectures in order to identify the most suitable model for achieving a high classification accuracy while maintaining robust generalization performance.

- Dmytro Dumanskyy:
- Tiago Kienzle Hagen: ResNet, CERN, Poster++
- Aleksandar Neshov: ResNet, EfficientNet, DenseNet
- Daniel Pfefferkorn: GoogLeNet, VGG-19, AlexNet

Maximizing the accuracy of the FER task represents the core goal of our group. Apart from the aforementioned aspects, the actual architecture of our neural network plays a key role in achieving this goal to the best of our ability. In the following section we want to highlight our decision making process.

2.1. Criteria for model selection

The suitability of a model for this project is evaluated based on the following criteria:

Firstly the design should be well documented in the literature to ensure scientifically proven effectiveness. For a

model to be taken into closer consideration it should have reported strong results on standard benchmarks.

Secondly the architecture should be capable of extracting important facial features such as the eyes, the eyebrows and the mouth meaningfully.

Finally models that are prone to overfitting are seen as unfavorable for our project. The chosen network should be able to learn distinctive features well while not overly relying on a high number of samples to achieve acceptable results. Since FER datasets are limited in sample size, we will prioritize models that shine through their data efficiency.

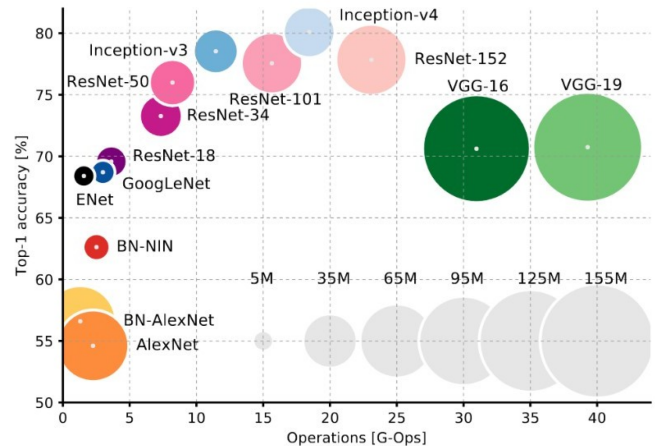


Figure 1. This figure from Kopalidis et al. [2] was used as a general reference as to which models to take into consideration.

2.2. Excluded model architectures

Not all types of models were categorized as suitable for the task at hand. Vision Transformers (ViT) are highly data hungry and typically rely on large-scale pretraining, making them suboptimal for training from scratch on limited FER datasets. Furthermore any outdated neural networks that are no longer competitive in terms of accuracy or efficiency were used as baselines for comparison.

2.3. Training setup

We decided to research common training parameters used in scientific papers that yielded good results. All models

are trained from scratch using a consistent training protocol to ensure fair comparison.

We decided to use Cross-Entropy loss as our loss function since it's widely used in similar classification tasks. Using Stochastic gradient descent (SGD) as our optimizer we are going to train the networks for approximately 30-50 epochs depending on individual model convergence. Furthermore we plan on exploring attention mechanisms to improve robustness against facial occlusions.

To ensure a consistent baseline for model evaluation all models are trained and tested on a predefined dataset.

2.4. Model selection strategy

We are currently in the process of deciding on a final neural network design for the entire group. Each team member is implementing their own model based on the architectures described above. The final model will be decided through the highest test accuracy on our collective test set.

3. Explainable AI

In order to portray the model's regions of interest in the image we plan on using GradCAM, which is a common method used for visualizing and interpreting networks by highlighting the image regions that contribute most strongly to a model's prediction.

References

- [1] Zhixin Dong, Chengdong Wu, Xiangyue Zhang, Xiaosheng Yu, and Junlin Li. A comparison of cnn-based 2d static facial emotion recognition techniques. *Journal of Physics: Conference Series*, 2832(1):012016, 2024. [1](#)
- [2] Thomas Kopalidis, Vassilios Solachidis, Nicholas Vretos, and Petros Daras. Advances in facial expression recognition: A survey of methods, benchmarks, models, and datasets. *Information*, 15(3), 2024. [1](#)